

Shambhawi Sharma (she/her)

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EDUCATION

M.Sc. Computer Science, **4.00**, University of Illinois at Chicago

Aug'24 - Dec'26

- Focus: Visualization, AI/ML, and NLP

B.Sc Computer Science, **3.91**, Summa cum Laude, Illinois Institute of Technology, Chicago, IL

Jan'21 - May'24

SKILLS

Languages: Python, C++, C, Java, Go, JavaScript, TypeScript, SQL, Shell Scripting

AI/ML: LDA, PCA, Machine Learning, Statistical Modeling, Psychometric Modeling, NumPy, MATLAB

Distributed Systems: OpenMPI, RPC, RMI, Sockets, Multithreading, Event-Driven Architecture, Virtualization

Backend & Cloud: REST APIs, Node.js, AWS, Azure, SQL, JSON

Tools: Git, GitHub, Jira, Unix/Linux, Docker, CI/CD

WORK EXPERIENCE

LCF Research Aide at Argonne National Laboratory

(Python | MATLAB | SQL | Data Analytics | Jira | Data Visualization | Git)

May '25 - Aug '25

- Developed STFT-based analysis pipelines for large-scale power, current, and GPU utilization time-series data at ALCF.
- Applied PCA to 5M+ spectrogram images to extract patterns in power consumption and GPU utilization across workloads.
- Analyzed GPU utilization and power across 5 supercomputing nodes to characterize workload-dependent performance.

Graduate Research at UIC, Illinois Tech, ALCF

(Python | C | C++ | Unix | Virtualization (Oracle) | OpenMPI | Git)

Jan '23 - Dec '26

- Developed ML perceptual models analyzing 1M+ data points across 2,000 dimensions to quantify human-computational perception differences; designed interactive Plotly, D3.js, and Vega-Lite systems to explore model outputs and perceptual patterns.
- Developed HPC scheduling algorithms for 2M+ jobs, reducing scheduling delays 15% and high-priority latency 20%.
- Analyzed distributed HPC workloads and DUMPI/OpenMPI event traces to optimize scheduling and resource allocation.

PROJECTS

Multi-Agent Research Orchestrator

(Python | asyncio | FastAPI | LLM APIs | Redis | PostgreSQL | Docker)

May '26 - Dec '26

- Built a multi-agent orchestration system decomposing research workflows into 50+ asynchronous subtasks across specialized supervisor/worker agents.
- Implemented 5+ reliability primitives including persistent memory, context summarization, task queues, retries, and failure recovery for long-running workflows.
- Designed 3+ tool-calling interfaces for web search, Python execution, and database queries, enabling dynamic tool selection and chaining.

Distributed P2P File Sharing System

(Python | Java | Multithreading | RMI | RPC | Sockets | Event-Driven Architecture)

Aug '24 - Dec '24

- Built a Gnutella-style distributed P2P system supporting 50+ nodes and 50+ simultaneous requests using RPC, sockets, and event-driven communication.
- Implemented push/pull consistency with configurable 1-5 minute TTR, reducing stale responses by 20% under concurrent workloads.
- Optimized query routing and lazy-pull synchronization, reducing query latency by 35% and bandwidth usage by 30%.

AI Search & Decision-Making Systems

(Python | NumPy | MCTS | GBFS | Heuristic Search | Backtracking)

May '25 - Dec '25

- Built AI solvers using MCTS, GBFS, MRV, and backtracking to evaluate alternative search and decision strategies.
- Optimized search-tree traversal and heuristic selection, achieving 70% faster solution times on complex problem instances.

Perceptual Noise Modeling for Visualization

(Python | NumPy | LDA | PCA | Statistical Modeling)

Aug '26 - Dec '26

- Extended an LDA-based perception model with Gaussian perceptual noise to simulate human-like decision uncertainty.
- Fitted noise parameters against human thresholds to quantify differences between computational and human perception.
- Modeled sensitivity across correlation levels using psychometric functions and quantitative threshold analysis.

POSTERS/PUBLICATIONS

- Reda K., **Sharma S.**, et al. Marks, Channels, and Dead Ends *BELIV '26*.
- Kurkure Y., **Sharma S.**, et al. CQSim+: Symbiotic Simulation for Multi-Resource Scheduling in HPC. *SIGSIM-PADS '25*.
- **Sharma S.**, et al On-Demand Pilot Job Scheduling for Adaptive Execution of HEP Workflows on HPC Systems. *GCASR '25*
- **Sharma S.**, et al Time-Frequency Characterization of GPU Power Behavior *CELS ANL '25*

ACHIEVEMENTS

- President & Treasurer, Vedic Vision Society | Dean's List | ACM & ACM-W | SIAM